

architecture and features that it has to offer.

Proprietary databases

Oracle and Microsoft SQL Server are not part of Drupal Core, but developers have built connectors to support them as databases for Drupal. These are proprietary systems and not usually chosen by open source architects, but the beauty of open source projects like Drupal is that they allow integration with any system and easily fit into a larger multi-technology stack architecture.

To learn more about the Oracle DB, log on to <https://www.drupal.org/project/oracle>, and for Microsoft SQL Server, you can go to <http://drupal.org/project/sqldr>

Caching mechanisms

The Drupal community has contributed many modules that split the Drupal caching mechanism from MySQL and other database options to object caching mechanisms like Memcache and Redis. So, all the cache tables like *cache*, *cache_page*, *cache_block* and more are hived off to the external caching mechanism.

Memcache

Memcache is a high-performance, distributed memory object caching system. It is an in-memory key-value store for small chunks of arbitrary data (strings, objects, etc) from results of database calls, API calls, or page rendering. PECL (the PHP Extension Community Library) provides for a Memcache or Memcached library that can be used with the Drupal module 'Memcache API and Integration' (<https://www.drupal.org/project/memcache>).

The module supports multiple memcached servers connected to the Drupal site, thereby distributing the load of the cache bins.

An example of the configuration that would go into Drupal's *settings.php* would be as below:

```
memcache_servers' => array(
  server1:port => cluster1,
  server2:port => cluster2,
  serverN:port => clusterN,
  'unix:///path/to/socket' => clusterS
)

'memcache_bins' => array(
  bin1 => cluster1,
  bin2 => cluster2,
  binN => clusterN,
  bins => clusterS
)
```

As a part of Acquia Cloud Enterprise, in case of high complexity and memory requirements, a dedicated server is assigned to host just the memcached services, while the

Apache2 Web server and the MySQL-Percona database server are dedicated to other instances.

Redis

Redis, too, is an advanced key-value cache and data store. It is often referred to as a data structure server. Drupal's module to support Redis as the cache store has been gaining popularity (<https://www.drupal.org/project/redis>).

The Redis Drupal module uses the Predis PHP library or PHPRedis PHP extension to provide connectivity to a Redis instance.

Entity cache

The Entity cache module is another very popular add-on to a Drupal site, which puts Drupal's core entities into cache via the cache API. This module, in combination with Memcache or Redis caching mechanisms, is potent. Core entities like Nodes, Taxonomy, Comments, Users and Files are offloaded from database queries and loaded into this Entity cache. One good thing is that, as of Drupal 8, this feature is being added to Drupal Core.

APC or Alternative PHP Cache

The APC module supports the Alternative PHP Cache (APC), which is a free and open source opcode cache for optimising PHP intermediate code. Along with the opcode cache, it also provides for a user cache for storing application data. APC can be used for cache data that does not change often and grows too big, to avoid fragmentation. The use of APC with other caching mechanisms can be combined by assigning which cache bin uses a particular cache cluster.

Boost

Boost is a file cache module that targets static page caching, especially in case of a Drupal site being hosted on a shared hosting provider. With respect to dedicated hosting, Varnish is a better option for page caching. Boost will cache and compress your HTML page, css, js, ajax and xml requests. It comes with an advanced cache expiration and regeneration logic that will surely speed up your site's performance for anonymous traffic.

Authcache

But what about authenticated users? The Authcache module comes to the rescue by offering page caching for both anonymous as well as authenticated users. It also integrates with other caching mechanisms like Memcache, Redis, MongoDB or any others that provide an implementation of DrupalCacheInterface. 

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